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Summer Semester, B.Tech. Course work
End-Semester Examination, June 2026
Course Title: *Probability and Stochastic Processes*
Course Code: CAMTC13/COMTC13

Time: 3 Hours Max. Marks: 50

Note: - Missing data/ information (if any), maybe suitably assumed & mentioned in the answer.

Q No	Attempt any two parts from each question	Marks	CO
Q 1			
1a	A cybersecurity firm employs three intrusion detection systems S_1, S_2, and S_3 to monitor network traffic. System S_1 analyzes 30% of the incoming packets, S_2 analyzes 45%, and S_3 analyzes the remaining packets. The probabilities that these systems incorrectly flag a legitimate packet as malicious are 0.02, 0.03, and 0.05, respectively. A packet selected at random is found to have been incorrectly flagged as malicious. Determine: (i) The probability that the packet was analyzed by system S_3. (ii) The probability that the packet was analyzed either by system S_1 or S_2.	5	CO1
1b	A smart grid controller records the number of power fluctuations occurring in a transmission line during a one-hour interval. The random variable X denoting the number of fluctuations has the following probability distribution: X: 0, 1, 2, 3, 4, 5 $P(X=x)$: 0.10, 0.15, 0.20, 0.25, 0.20, 0.10 The cost incurred (in dollars) due to x fluctuations is given by $$C(X) = X^2 + 2X + 5.$$ Determine the expected number of fluctuations, the variance of the number of fluctuations, and the expected cost incurred during a one-hour interval.	5	CO1
1c	The number of packets transmitted successfully in a communication network during a fixed interval is represented by a random variable X. The first four moments about the origin are given as $$\mu_1' = 4, \quad \mu_2' = 20, \quad \mu_3' = 112, \quad \mu_4' = 676.$$ Determine the variance of X, the third and fourth moments about the mean, and the coefficient of kurtosis. Comment on the nature of the distribution.	5	CO1
Q 2			
2a	A network administrator recorded the number of packet losses occurring per minute in a communication channel during 200 observation periods. The data obtained are as follows: Number of packet losses (x): 0, 1, 2, 3, 4, 5 Frequency (f): 30, 52, 60, 34, 18, 6 Fit a Poisson distribution to the above data and obtain the expected frequencies.	5	CO2
2b	The arrival time X (in minutes) of a maintenance engineer at a communication hub is assumed to be uniformly distributed over the interval $[10, 30]$. Determine: (i) The probability density function of X. (ii) The probability that the engineer arrives between 15 and 25 minutes. (iii) The mean arrival time. (iv) The variance of the arrival time.	5	CO2
2c	A cloud computing service provider observes that the response time X (in milliseconds) of a server is a non-negative random variable with mean response time equal to 120 ms. Using Markov's inequality, obtain an upper bound for the probability that the response time exceeds 300 ms. Further, determine an upper bound for the probability that the response time exceeds 480 ms.	5	CO2
Q 3			
3a	The random variable X has p.m.f.: $$f(x) = \begin{cases} \dfrac{e^{-\theta}\theta^x}{x!}, & x = 0, 1, 2, \ldots, \\ 0, & \text{elsewhere.} \end{cases}$$ (i) Find its m.g.f., (ii) Using part (i), find the m.g.f. of $Y = 2X - 1$.	5	CO3

3b	(i) State Central Limit Theorem. (ii) A die is thron 600 times. Using the Central Limit Theorem, find the approximate probability that the number of sixes obtained lies between 80 and 120. Given $P(0 \leq Z \leq 2.19) = 0.4857$.	5	CO3
3c	Fit the regression line of Y on X for the following data using the method of least squares and estimate the blood pressure when the age is 45 years:	5	CO3

Age (X)	56	42	72	36	63	47
Blood Pressure (Y)	147	125	160	118	149	128

Q 4			
4a	A bank manager claims that the average age of customers who open savings accounts through his branch is less than the average age of customers of all branches, which is known to be 35 years. To verify this claim, a random sample of 100 customers who opened accounts through his branch was selected. Their age distribution is given below:	5	CO4

Age (years)	Number of Customers
20–24	14
25–29	24
30–34	28
35–39	20
40–44	14

Test the manager's claim at the 5% level of significance.($P(Z < 1.645) = 0.95$.)

4b	The average lifetime of a certain type of battery is claimed to be 1200 hours. The population standard deviation is known to be 120 hours. A sample of 144 batteries has an average lifetime of $\bar{X} = 1170$ hours. The null hypothesis is rejected if the sample mean is less than 1183.55 hours. Assume $H_0 : \mu = 1200$ and $H_1 : \mu < 1200$. Find the 95% confidence limits for the true mean battery life, Type I Error, and Type II Error when the true mean lifetime is $\mu = 1160$ hours. Given $P(Z < 1.645) = 0.95$ and $P(Z < 2.355) \approx 0.9907$.	5	CO4

4c	A call center receives customer complaints throughout the week. During a particular week a call center recorded a total of 134 customer complaints. The number of complaints received on Sunday, Monday, Tuesday, Wednesday, Thursday, Friday, and Saturday were 18, 24, 15, 21, 17, 20, and 19, respectively. Examine whether the complaints are uniformly distributed across the days of the week using the Chi-square goodness-of-fit test at the 5% level of significance. Given:$\chi^2_{6,0.05} = 12.59$ and $\chi^2_{7,0.05} = 14.067$.	5	CO4
Q 5			
5a	Two brands of LED bulbs, Brand A and Brand B, were tested for their service life. Independent random samples yielded the following results:	5	CO5

Particulars	Brand A	Brand B
Sample Size	$n_1 = 12$	$n_2 = 10$
Sample Mean Life	$\bar{x}_1 = 1520$ hours	$\bar{x}_2 = 1450$ hours
Sample Variance	$s_1^2 = 900$	$s_2^2 = 400$

Assuming that the lifetimes are normally distributed:
(i) Test at the 5% level of significance whether the population variances of the two brands are equal.
(ii) If the variances may be regarded as equal, test whether Brand A has a significantly greater mean lifetime than Brand B. ($F_{11,9}(0.05) = 3.10$ and $t_{20,0.05} = 1.725$).

5b	Explain and discuss the properties of a good estimator.	5	CO5
5c	Derive the relationship between t and F distribution.	5	CO5